k-Anonymity-Based Data Anonymization and Re-identification Risk Analysis

MSCS 714 Data Privacy Seminar

Group Project

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Problem

- Removing direct identifiers is not enough to prevent re-identification.
- Quasi-identifiers such as age, sex, education, occupation, and country can still enable linkage.
- We study how k-anonymity changes exact-match linkage risk and data utility.

Dataset

- Adult Census Income dataset from UCI
- 32,561 rows and 15 columns
- Sensitive attribute: income (>50K vs <=50K)
- QI Set A: age, sex, education, marital status
- QI Set B: age, sex, occupation, native country

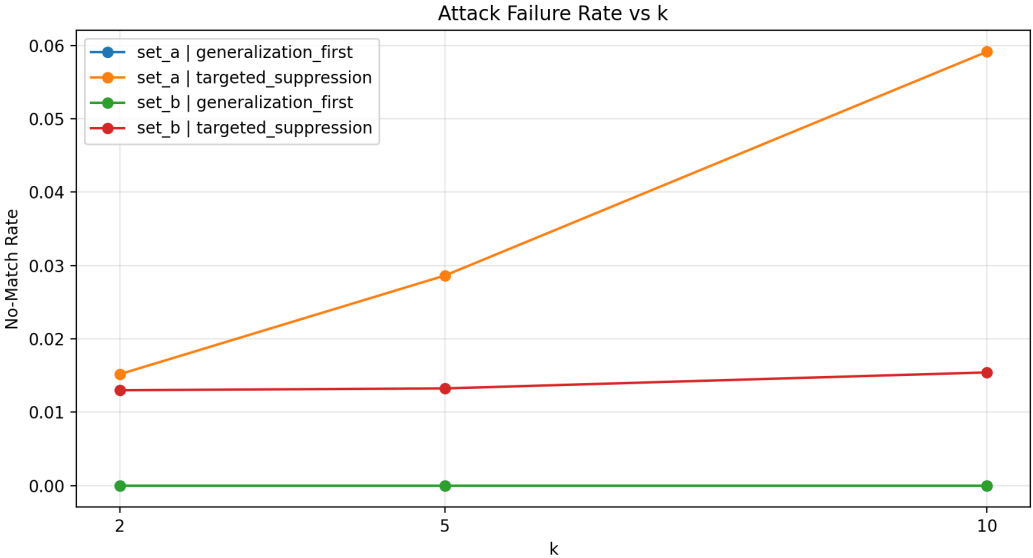
Method

- Privacy levels: $k = 2, 5, 10$
- Strategies:
  - generalization-first
  - targeted suppression with mild generalization
- Linkage attack: exact matching with five auxiliary samples
- Utility metrics: retention, income shift, hours-per-week distortion, information-loss score

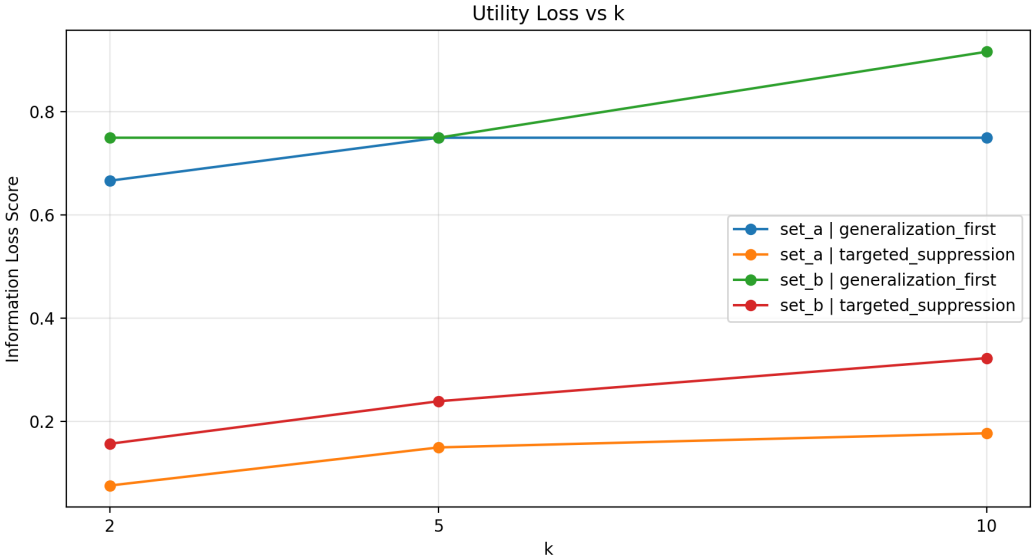
Validation

- All 12 generated releases pass explicit k-anonymity validation.
- Automated tests pass on the pipeline.
- Unique match rate becomes zero in every evaluated setting.

Attack Failure vs k



Utility Loss vs k



Key Results

- Generalization-first keeps all rows but has very high information loss.
- Targeted suppression keeps 94.0% to 98.7% of rows depending on the setup.
- QI Set A with targeted suppression at $k = 2$ gives the best balance in this implementation.
- Stronger privacy increases attack failure, but also increases information loss.

Example Comparison

Setup	Retention	No-Match	Info Loss	Income Shift
Set A, TS, $k = 2$	98.6%	1.5%	0.076	0.0014
Set A, TS, $k = 10$	94.0%	5.9%	0.177	0.0065
Set B, TS, $k = 2$	98.7%	1.3%	0.157	0.0004
Set B, GF, $k = 10$	100.0%	0.0%	0.917	0.0000

Limitations

- This project evaluates only k-anonymity, not l-diversity or t-closeness.
- The attacker is simulated from the same source population.
- Utility is measured with lightweight descriptive metrics rather than full downstream tasks.

Conclusion

- The project delivers a full k-anonymity workflow with validation and reporting.
- Exact unique linkage can be eliminated, but the privacy–utility trade-off remains central.
- The most practical release in this implementation is QI Set A with targeted suppression at low k.

Demo Flow

- Open the repository and show the dataset structure.
- Run `python scripts/run_experiments.py`.
- Show `results/metrics/experiment_summary.csv`.
- Show the two generated plots.
- Open the final report PDF.